

Customer Sales Analysis Project

1. Project Overview and Objectives

Project Overview

The Customer Sales Analysis project is an end-to-end data analytics pipeline that integrates sales transactions with customer churn data to uncover purchasing patterns and retention insights. It applies data cleaning, feature engineering, aggregation, and visualization to identify high-value customers, sales trends, and cross-selling opportunities, supporting data-driven business decisions.

Objectives

- To build an end-to-end customer sales analytics workflow using Python and pandas
- To clean, validate, and integrate sales and churn datasets
- To analyze customer behavior, regional sales, and product performance
- To identify churn patterns and cross-selling opportunities
- To create professional visualizations and generate business-ready insights

2. Setup and Installation Instructions

Prerequisites

- Python 3.10+
- Jupyter Notebook

Installation Steps

1. Download or clone the project repository from GitHub.
2. Install required libraries:

```
pip install -r requirements.txt
```
3. Place dataset inside /data folder.
4. Launch Notebook:

python notebook

5. Open customer_analysis.ipynb.

3. Code Structure Explanation

```
customer-sales-analysis_week5/  
|  
├── data/  
|   ├── sales_data.csv  
|   └── customer_churn.csv  
|  
├── notebooks/  
|   └── customer_analysis.ipynb  
|  
├── reports/  
|   └── analysis_report.pdf  
|  
├── requirements.txt  
├── README.md  
└── .gitignore
```

File Descriptions

- notebooks/customer_analysis.ipynb
Main notebook performing data loading, cleaning, merging, analysis, visualization, churn evaluation, cross-selling, and business insights.
- data/sales_data.csv
Sales transactions with dates, products, quantities, prices, customer IDs, regions, and total sales.

- data/customer_churn.csv
Customer details including tenure, billing, contract type, payment method, and churn status.
- reports/analysis_report.pdf
Final analytical report summarizing key metrics, visualizations, insights, and business recommendations.
- notebooks/.ipynb_checkpoints/
Automatically generated Jupyter backup folder.
- requirements.txt
List of Python libraries required to run the project.
- README.md
Project documentation including overview, setup instructions, and usage guide.

4. Screenshots of Working Application

The included screenshots demonstrate:

- Both sales and customer churn datasets were loaded successfully
- Monthly expense trends were visualized correctly
- Region-wise and product-wise sales were computed and displayed
- Customer churn distribution and lifetime value were visualized using charts
- Cross-selling analysis and pivot tables were generated without errors

```
C:\Users\N...>python -m notebook
[1 2026-02-13 18:35:17.140 ServerApp] jupyter_lsp | extension was successfully linked.
[1 2026-02-13 18:35:17.149 ServerApp] jupyter_server_terminals | extension was successfully linked.
[1 2026-02-13 18:35:17.165 ServerApp] jupyterlab | extension was successfully linked.
[1 2026-02-13 18:35:17.179 ServerApp] notebook | extension was successfully linked.
[1 2026-02-13 18:35:18.593 ServerApp] notebook_shim | extension was successfully linked.
[1 2026-02-13 18:35:18.721 ServerApp] notebook_shim | extension was successfully loaded.

To access the server, open this file in a browser:
file:///C:/Users/N.../AppData/Roaming/jupyter/runtime/jpserver-26248-open.html
Or copy and paste one of these URLs:
http://localhost:8888/tree?token=34216f0a01eca616beed8564920098d97181e4e31a11a001
http://127.0.0.1:8888/tree?token=34216f0a01eca616beed8564920098d97181e4e31a11a001
```

Fig.4.1. Terminal Output

- Final reports and visualizations executed successfully within Jupyter Notebook

Product	Headphones	Laptop	Monitor	Phone	Tablet
Region					
East	288361	221946	642870	506828	859634
North	107091	1798206	397100	489284	1191954
South	512168	1373120	39924	1471428	341212
West	476413	495938	268177	391854	491540

Fig.4.1. Pivot Table

Product	Headphones	Laptop	Monitor	Phone	Tablet
Product					
Headphones	1.000000	-0.236067	-0.176471	-0.210042	-0.249004
Laptop	-0.236067	1.000000	-0.236067	-0.280976	-0.333096
Monitor	-0.176471	-0.236067	1.000000	-0.210042	-0.249004
Phone	-0.210042	-0.280976	-0.210042	1.000000	-0.296374
Tablet	-0.249004	-0.333096	-0.249004	-0.296374	1.000000

Fig.4.2. Cross Selling (Correlation Matrix)

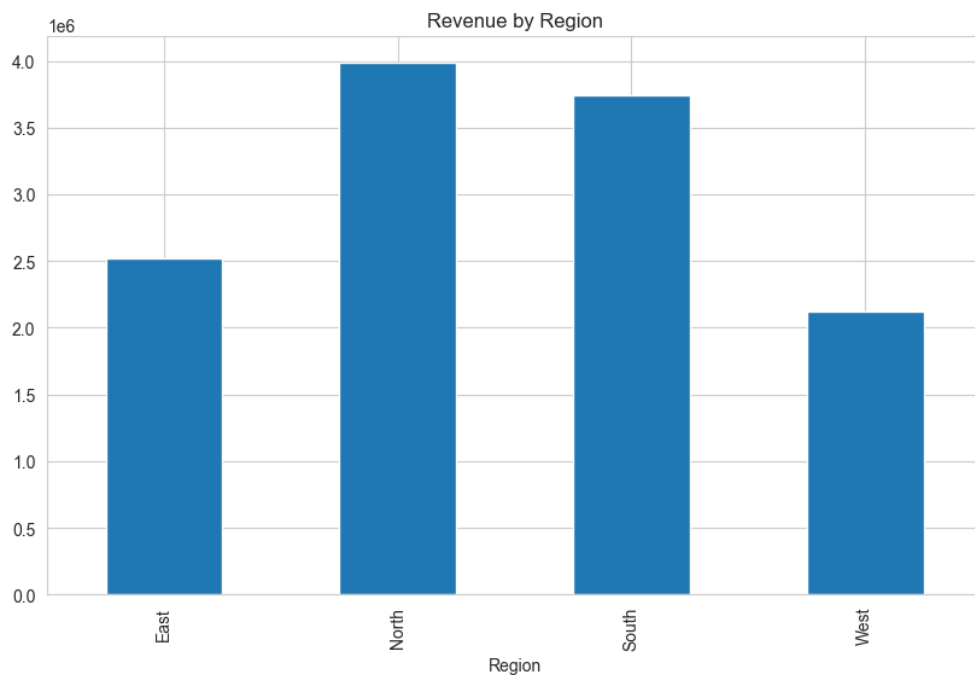


Fig.4.3. Revenue by Region

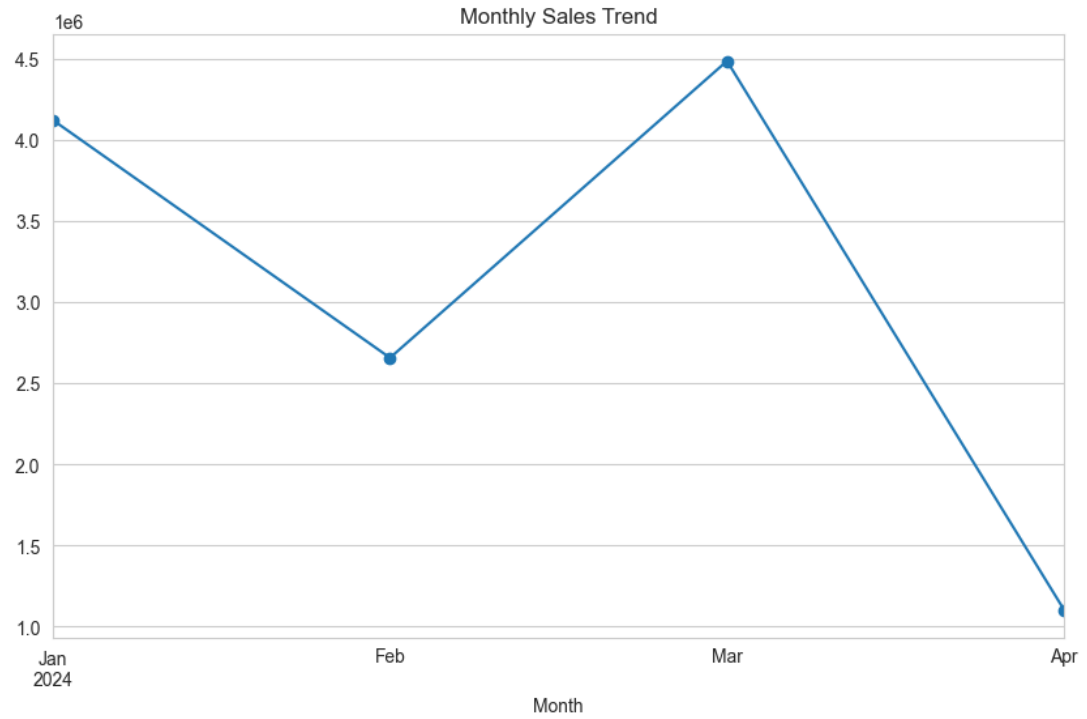


Fig.4.4. Monthly Sales Trend

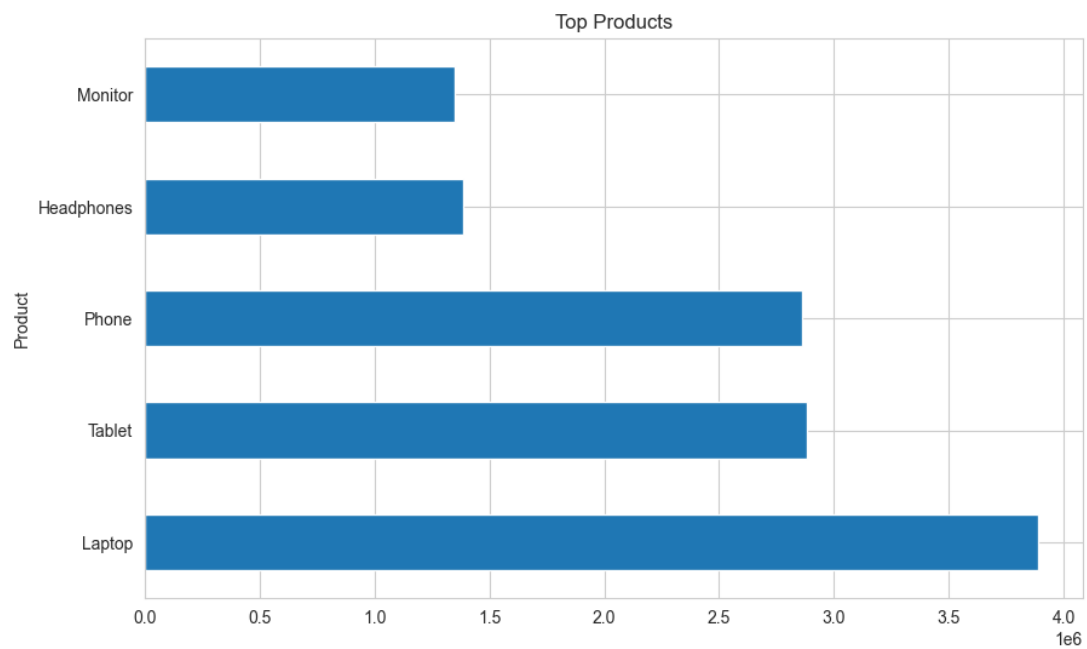


Fig.4.5. Horizontal Bar Chart for Top Products

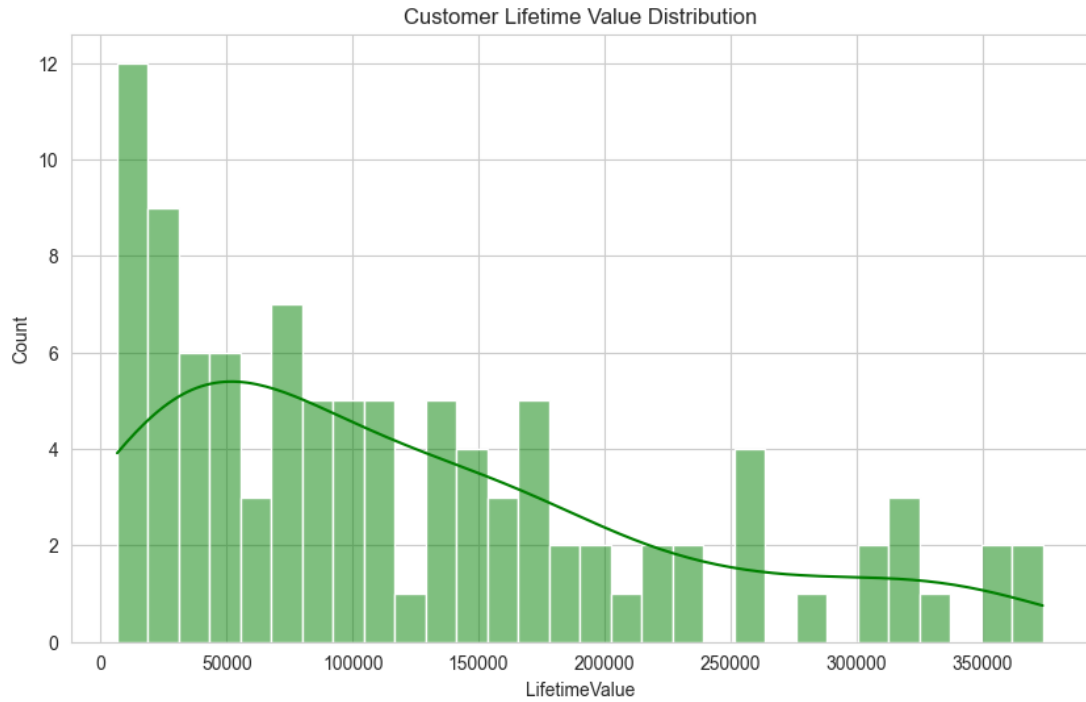


Fig.4.6. Customer Lifetime Value Distribution

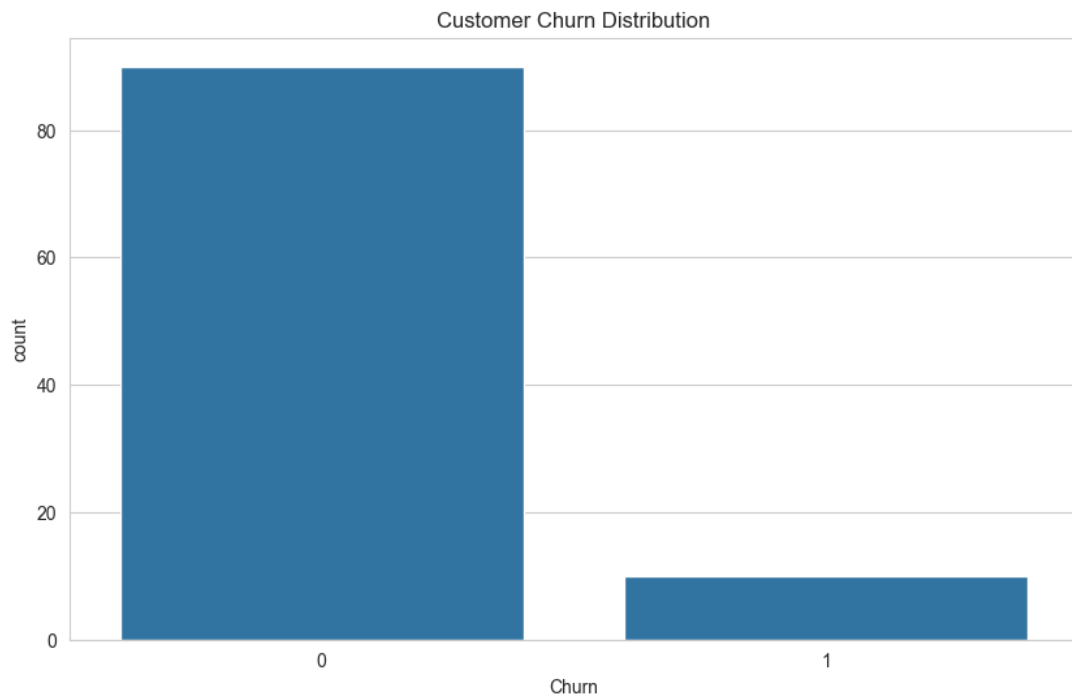


Fig.4.7. Customer Churn Distribution

This visual proof confirms that the application runs correctly and satisfies all functional requirements.

5. Technical Requirements Fulfillment Explanation

Category	Technical Requirement	How It Was Implemented
Algorithm	Data loading and preprocessing flow	A sequential pipeline was implemented to load sales and churn CSV files, validate columns, normalize customer IDs, clean the datasets, and prepare them for analysis using pandas.
Algorithm	Data cleaning logic	Missing numerical values were handled using median imputation, dates were converted to datetime format, invalid values were coerced to NaN, and datatypes were standardized.
Algorithm	Sales metric calculations	Aggregation functions such as sum(), mean(), nunique(), and groupby() were used to compute total revenue, average order value, regional sales, and product performance.
Algorithm	Trend identification	Monthly sales trends were identified using to_period() grouping and visualized through line charts for time-based analysis.
Algorithm	Customer lifetime value computation	Customer Lifetime Value was calculated by grouping total sales per customer and merged back into the main dataset for advanced analysis.
Algorithm	Cross-selling analysis	A customer-product matrix was created using groupby() and unstack(), converted to binary format, and correlation analysis was applied to identify frequently purchased product combinations.
Data Structure	Tabular data handling	Pandas DataFrames were used to efficiently manage and manipulate structured sales and customer churn data.
Data Structure	Metric storage	Aggregated metrics such as revenue, churn rate, and lifetime value were stored using Series and DataFrames for organized access and reporting.

Category	Technical Requirement	How It Was Implemented
Architecture	Dataset integration	Sales and churn datasets were merged using normalized customer identifiers with an inner join to ensure consistent records.
Architecture	Modular notebook design	The notebook was organized into logical sections including loading, cleaning, feature engineering, aggregation, visualization, and insight generation.
Architecture	Configuration management	Visualization settings and display options were defined at the beginning of the notebook for centralized control.
Output	Automated analytics dashboard	Multiple charts (regional sales, monthly trends, top products, churn distribution, lifetime value) were generated automatically using Matplotlib and Seaborn.
Output	Business reporting	Key metrics, pivot tables, and analytical insights were produced directly within the notebook and summarized in the final project report.

6. Learning Outcomes

Through this project, the following learning outcomes were achieved:

- Gained hands-on experience in building an end-to-end customer sales analytics pipeline using Python and pandas.
- Learned to clean, validate, and integrate multiple real-world datasets, including handling missing values and date normalization
- Developed skills in calculating business KPIs such as revenue, churn rate, and customer lifetime value.
- Improved proficiency in data visualization using Matplotlib and Seaborn for trend and distribution analysis.

- Strengthened understanding of structured data analysis, feature engineering, and exploratory analytics for business decision-making.

7. Conclusion

This project successfully implemented a complete customer sales analysis workflow, integrating transactional and churn data to extract meaningful business insights. Key metrics, trends, and customer behaviors were identified through structured analysis and visualization, demonstrating practical data analytics skills applicable to real-world business scenarios.